

Project Presentation of Industry Oriented Hands-on Experience (22CS422) On **PathWise**

AI-Based Academic and Career Recommendation System for Students

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INTRODUCTION



Choosing the right academic or career path is one of the most important decisions for students. However, most students struggle due to lack of personalised guidance, unclear career information and absence of a structured plan.

PathWise is an **AI-driven** platform that helps students to discover career paths that best match their skills, interests and goals. It uses advanced algorithms to provide personalised career recommendations, identify skill gaps, and generate step-by-step learning roadmaps.



Provides accurate and **personalized** career recommendations



Identifies **skill gaps** and suggests improvements



Offers structured **learning roadmaps** with resources



Ensures **explainable** and **transparent** recommendations



Empowers students to make **confident** career decisions



PROBLEM STATEMENT



Students struggle to choose the right academic or career path due to lack of **personalized, explainable, and structured** guidance, leading to **poor academic and career decisions**.

KEY CHALLENGES



No Personalization

Generic advice for everyone



Lack of Guidance

Unclear direction and support



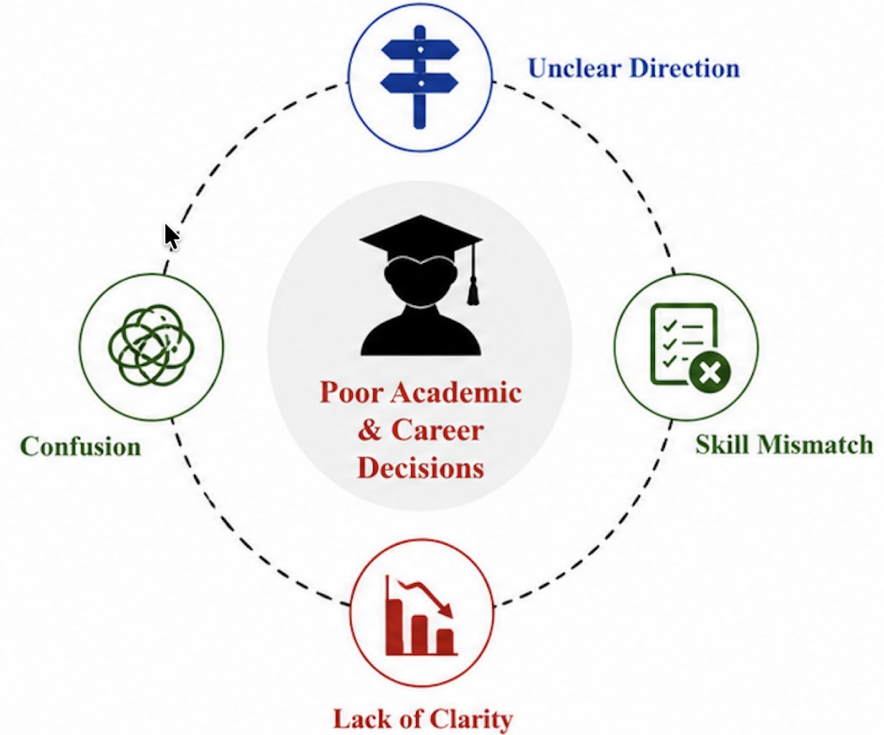
Skill Gaps

Students don't know what skills to improve



Unstructured Information

Scattered information across platforms



An intelligent system is needed to provide **personalized** recommendations, identify **skill gaps**, and offer **structured learning roadmaps**.

OBJECTIVE



The main objective of **PathWise** is to build an intelligent and reliable platform that guides students towards the right **academic** and **career** path.



1. Provide Personalized Career Recommendations

Analyze skills, interests and goals to suggest suitable career options.



2. Identify Skill Gaps

Detect the skills required for chosen careers that the user lacks.



3. Generate Learning Roadmaps

Offer structured, step-by-step roadmaps with courses, certifications and projects.



4. Enable Resume Parsing

Extract skills automatically from uploaded resumes to improve accuracy.



5. Provide Explainable Recommendations

Show match scores and visual insights for transparent suggestions.



6. Ensure Secure Authentication

Implement secure login, OTP verification and data protection.



7. Conduct Skill Assessments

Offer practice tests to evaluate user's skills and knowledge.



8. Implement Anti-Cheating Mechanism

Detect malpractice during tests to ensure assessment integrity.



9. Collect User Feedback

Improve system continuously based on user feedback.

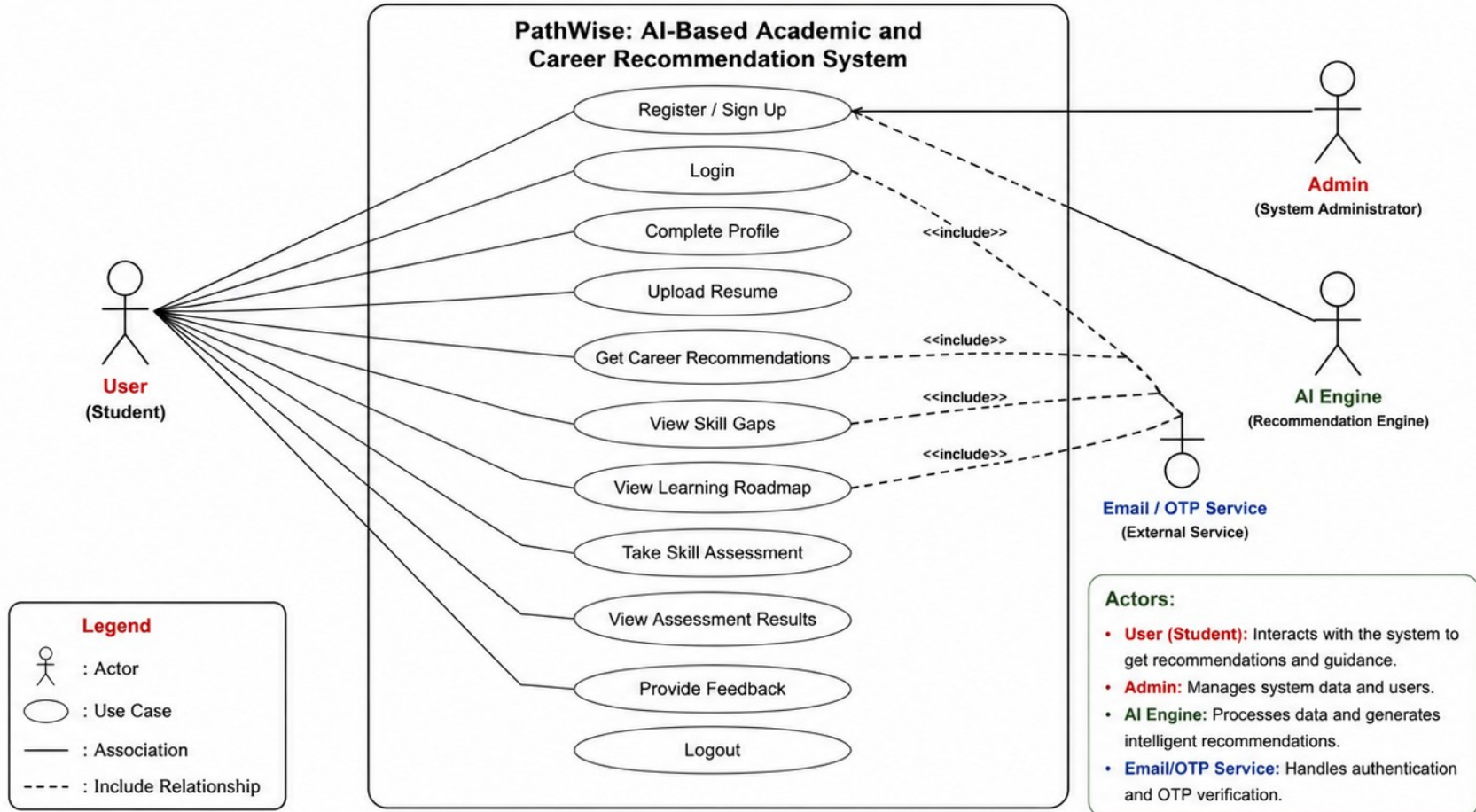


PathWise aims to empower students with **data-driven insights**, helping them make confident, **informed** and **future-ready** academic and career decisions.

UML DIAGRAM



Use Case Diagram



TOOLS & TECHNOLOGIES



PathWise is built using modern, **scalable**, and **secure** technologies to deliver **intelligent** and **personalized** career recommendations.

FRONTEND



- Next.js
- TypeScript
- Tailwind CSS
- Recharts
(Data Visualization)

BACKEND



- Node.js
- Express.js
- RESTful APIs

DATABASE



- MongoDB
- Mongoose ODM

AI ENGINE



- Node.js
- TypeScript
- Cosine Similarity
- Weighted Scoring
- Rule-Based Filtering

AUTHENTICATION & SECURITY



- JWT (JSON Web Tokens)
- bcrypt (Password Hashing)
- OTP Verification
- Secure Session Management

FILE & DATA PROCESSING



- Multer (File Upload)
- PDF Parser
- Resume Parsing
- Data Preprocessing

TESTING & ASSESSMENT



- MCQ Based Tests
- Anti-Cheating System
- Tab Switch Detection
- Result Evaluation

DEPLOYMENT & TOOLS



- Vercel (Frontend Deploy)
- Render (Backend Deploy)
- Git & GitHub
- Postman (API Testing)



These technologies work together to ensure a **fast**, **secure**, **scalable**, and **user-friendly** experience.

SNAPSHOTS FROM THE PROJECT

CHITKARA
UNIVERSITY



PathWise

Dashboard

Profile

Recommendations

Practice Tests

AI Recommendation Engine

Uses similarity + skills + interests to generate personalized career paths.

Jinny Kapur

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Logout

N

PathWise v1.0 | AI Career System

Dashboard

AI-powered career insights based on your profile.

Refresh

View Recommendations →

Top Match

Based on similarity, skills & interests

55%

Full-Stack Web Development

Similarity

47%

Skills

83%

Interests

33%

Profile Snapshot

Used by AI engine

Skills

Python

Java

C

C++

JavaScript

HTML

CSS

React

Interests

Design

Systems

Data

Recommendations are based on skills & interests.

Edit profile →

Next Steps

Follow these actions to improve your career match score.

Complete Profile

Add skills & interests to improve AI accuracy.

Update Profile →

Fix Skill Gaps

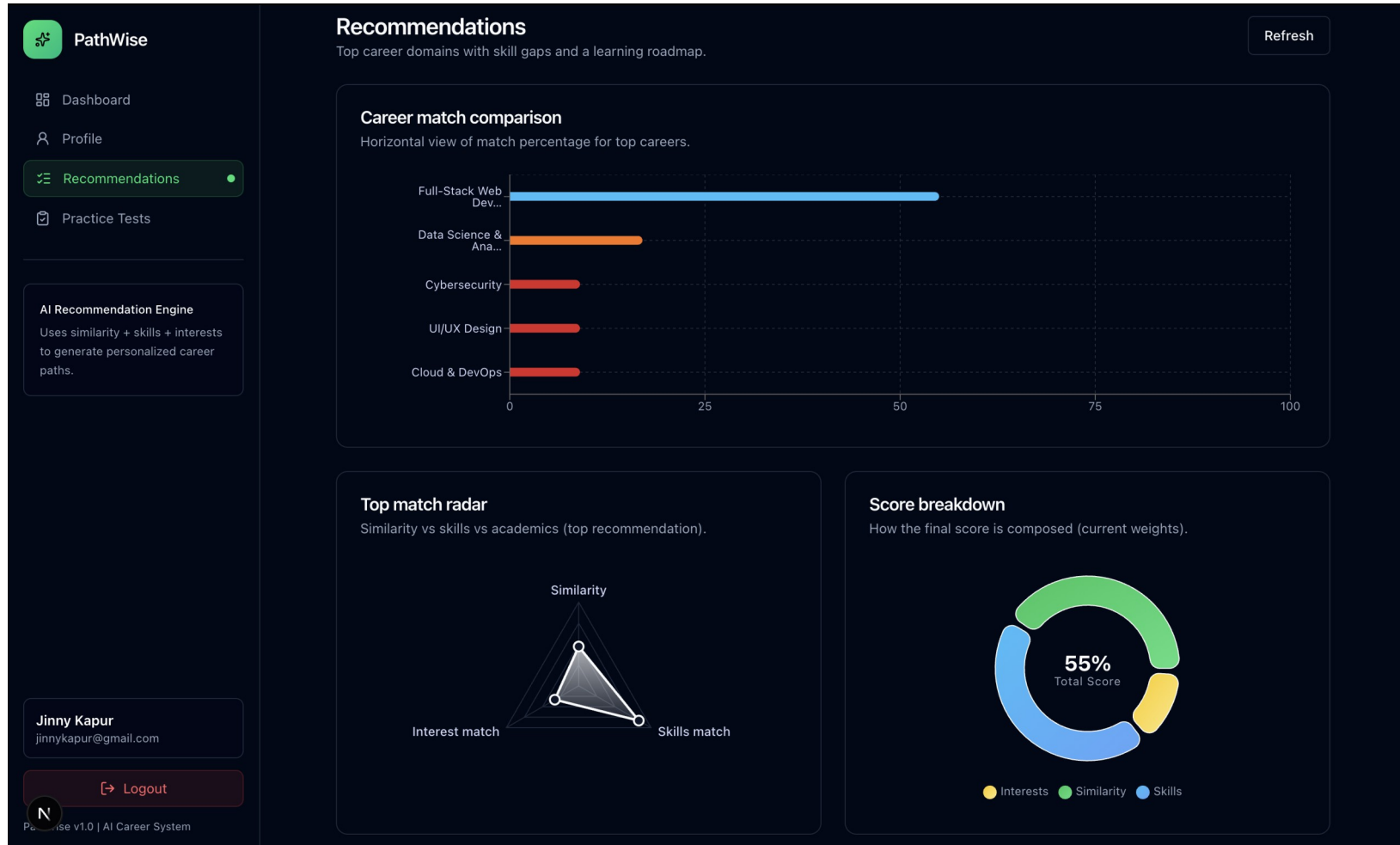
Focus on missing skills for your top career.

View Gaps →

Take Skill Test

Validate your skills with AI-based assessment.

Start Test →



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PathWise v1.0 | AI Career System

Skill gaps (top match)

Matched vs missing skills count for the top recommendation.

Skill Coverage

5 / 6

Matched Skills

5

Missing Skills

1

Readiness Score

83%

Based on current skill match vs required skills

Missing skills to focus

typescript

Full-Stack Web Development

47% similarity · 83% skills · 33% interest

55%

Matched Skills

javascript react node.js mongodb rest apis

Skill Gaps (Focus Here)

typescript

Matched Interests

design

Learning Roadmap

1 Close skill gaps: typescript

2 Take courses: Full-Stack Open · React + TypeScript

3 Earn certifications: Meta Front-End Developer

4 Build 2–3 relevant projects aligned to this domain and document them well.

 PathWise

 Dashboard

 Profile

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PathWise v1.0 | AI Career System

Full-Stack Web Development Roadmap

Step-by-step learning path from beginner to advanced.

Back

Practice or Skill Test

Refresh

Progress

71%

5 completed7 total

Level completed. You can now practice your skills in the test section.

● Beginner

Completed

✓ skill TypeScript

Open skill test in new window

✓ course Full-Stack-Open

University of Helsinki

✓ course React + TypeScript

Udemy

✓ project Portfolio website (responsive + dark mode)

● Intermediate

Completed

✓ project Full-stack CRUD app (auth + database + API)

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Skill Test Window

Read instructions first, acknowledge integrity rules, then start the timed test.

[Back to practice launcher](#)

Test Instructions and Integrity Acknowledgement

This test starts only after you acknowledge the statements below.

1. This is a self skill test for personal growth and planning.
2. The test has 10 MCQ questions and a 15-minute timer.
3. Do not use external help, tab switching, or copied answers during the test.
4. Repeated switching events add warnings and can flag/auto-submit your attempt.

Selected skill

Java

- ☒ I confirm that I am genuinely giving this test for self skill evaluation.
- ☒ I confirm that I will not cheat during this test.
- ☒ I confirm that all answers I submit will be my own.

Proceed and Start Timer

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 **N**
PathWise v1.0 | AI Career System

Skill Test Window

Read instructions first, acknowledge integrity rules, then start the timed test.

[Back to practice launcher](#)

Java timed skill test

Warnings: 2

Time left: 08:17

Warning: repeated tab/window switching can auto-submit and flag this attempt.

10 / 10 answered

Marks: 10

Q1. For intermediate Java, what most improves reliability in problem solving?

Java

- ☐ Defer all refactoring indefinitely
- ☐ Hardcode values for speed
- ☐ Rely on manual checks only
- ☒ Use repeatable workflows, tests, and observability

Q2. You are optimizing debugging in Java. Which decision is strongest?

Java

- ☒ Measure baseline, apply targeted optimization, re-measure
- ☐ Optimize every file equally
- ☐ Pick changes without measurement

CONCLUSION



PathWise achieves its goal of delivering **intelligent** and **personalized** academic and career guidance using **AI**.

The system helps students discover the right career path, identify skill gaps, and follow a structured roadmap towards their goals.



Achievement: Successfully delivers accurate career recommendations and skill gap analysis.



Impact: Empowers students to make informed academic and career decisions with confidence.



Benefits: Saves time, reduces confusion, and focuses students on the right skills and opportunities.



Future Scope: Can be enhanced with more data, advanced AI models, and industry partnerships for greater accuracy and reach.



In conclusion, PathWise is more than just a recommendation system — it is a **smart companion** that **guides** students **today** for a **better** and **successful** tomorrow.

FUTURE SCOPE



PathWise lays a strong foundation for AI-driven academic and career guidance. In the future, it can be further enhanced to deliver more **powerful**, **accurate**, and **impactful** recommendations.



Advanced AI Models

Integrate more advanced machine learning models and deep learning techniques for higher accuracy.



More Data & Personalization

Use larger datasets and real-time data to provide more personalized and adaptive recommendations.



Industry Collaboration

Partner with industries and universities to align recommendations with current job market trends.



Mobile Application

Develop a mobile app to make PathWise accessible anytime, anywhere.



Continuous Improvement

Implement user feedback loops and performance analytics to keep improving the system continuously.



The future of PathWise is about continuous **innovation**, **collaboration**, and **expansion** to empower every student with the best guidance for a **successful** tomorrow.

THANK YOU!

